

Sustainability and Energy

When referring to sustainability regarding energy use, one of the factors that must be looked into is sustainable energy, befitting the growing human population. Since the industrial era, people have been using coal, and later on diesel oil, to fuel their machinery, ranging from manufacturing products to powering electrical units to fueling cars. For decades, this has been the main fuel source almost globally, but how sustainable truly are these fossil fuels?

As the name implies, fossil fuels originate from fossils - or carcasses - formed millions of years ago, and because humanity cannot wait a couple million years each time we are in need of energy, they are considered a non-renewable resource. However, it's not just its non-renewability that makes it so unsustainable.

With the skyrocket of fossil fuel use since the 1950s, so have yearly carbon dioxide emissions from these fuels exponentially increased in turn. As a greenhouse gas, carbon dioxide reflects radiated heat back into the atmosphere, ensuring the planet remains at a habitable temperature; however, with the increase in its emissions, global temperatures have increased, causing worse droughts in dry regions and flooding along the coastline as ice caps begin to melt at the poles.

Combining all of that, at the rate the planet is warming, if things keep going as they are, it will cause national climate disruption and may end with catastrophic consequences leading to the loss of lives and struggles in agricultural production. All of this to say, renewable or not, fossil fuels need replacement.

Luckily, we do have options. Those being nuclear power, solar power, wind power, and geothermal energy, and wave power. Let us turn to the most promising long-term solution. Nuclear power.

As defined in the wiki, nuclear power is the use of nuclear reactions to generate electricity, reactions such as nuclear decay, nuclear fusion, and nuclear fission. We'll focus on nuclear fission as of now, seeing as it is the most widely used method of generating power at nuclear power plants.

Uranium and plutonium are primarily utilized in this reaction due to their key roles in nuclear energy production. These elements offer significant advantages as safe and clean energy resources, capable of supplying power for many millennia thanks to the vast abundance of their ores and their long-lasting properties. Their use not only helps meet our steadily increasing energy demands but also contributes to the reduction of harmful carbon emissions, making nuclear power a vital component of a sustainable energy future.

One may ask how nuclear energy is to be used. Well, nuclear fission occurs in a high-pressure reaction chamber, where large unstable nuclei decay into smaller, more stable ones by breaking down and emitting neutrons. These stray neutrons hit the smaller nuclei just formed and cause

them to decay. This makes a chain reaction that releases high amounts of energy, depending on the scale.

As for nuclear fusion, that's harder for us to get a hold of. It's the same process that takes place in stars, providing sufficient energy for the star to continue existing. To put it simply, nuclear fusion can be described as the joining of two smaller nuclei into a bigger one, releasing much more energy than nuclear fission. We have tried to replicate this in special chambers called tokamaks. However, we have only partially succeeded. Scientists all around the world have been trying to build a tokamak that functions by itself, or at least yields an energy gain. This has been achieved several times throughout history, but only for short bursts of time. Instances of this in action include: NIF in the USA, JET in the UK, and EAST in China.

Nuclear power portrays some of the lowest causes of fatality yearly, and one study suggested that with each power plant built, up to 800,000 individuals would be averted from air pollution-related deaths caused by fossil fuels. The energy released by both forms of nuclear power can be calculated by Albert Einstein's famous equation $E = mc^2$.

With all this said, it does not come without its downsides. Even though it generates some of the greatest amounts of clean energy yearly, it also requires a lot of investment to maintain and ensure it remains safe, as the radiation it causes must be handled with precision and care. This leads us to other options, mainly things such as solar panels, wind turbines, and hydroelectric generators, which are forms of renewable energy. These options have become much more widespread in recent times, helping reduce their costs over the past decade or so.

We'll begin with solar panels. They use a range of technologies to absorb the light and heat the sun emits and transform it into usable electricity. Solar panels have become relatively affordable and in places that they can be placed in bulk, such as deserts and recently, forests in concave tree shapes so as to not disrupt the ecosystem nor tear down trees unnecessarily, their energy output has only increased as the years have gone by, and while alone they are barely able to generate enough electricity to power a house, depending on the model and size, they have only gotten more and more efficient and upon being used in large amounts, become a reliable source in open, sunny places. However, they are much less useful in areas that are cloudy, have short daytime periods, or have unpredictable weather, making them not the most reliable source of energy due to the lack of sun.

This brings us over to wind turbines. A wind turbine is a device that converts the kinetic energy that wind generates into electrical energy. These same wind turbines tend to be placed one beside each other in grand scales called wind farms in order to generate as much energy as possible due to the fact that, similar to solar panels, they do not produce much electricity alone. They are very alike in that they both require carefully thought-out decisions to ensure that the most is made out of them, which is why the information regarding the exact geographical location that would generate the most wind for these turbines is the most crucial hurdle to get your hands on. Unfortunately, these turbines also tend to take up a lot of space that could be used in other,

slightly more useful, ways, leading to these investments often getting pushback in favor of agricultural space or housing projects. On the other hand, they are one of the least cost heavy equipment to manufacture, thus making it easy to make them at a rather large scale, meaning it depends entirely on the space available within a country.

Moving on, it's time to look at hydroelectricity. Hydroelectricity is the renewable energy source that creates the greatest percentage of all clean global energy sources, making more than all the other renewable energy sources combined. It can also provide us with a large amount of energy on demand through using water reservoirs. However, due to using dams more often than not, it has its own fair share of issues, like everything else. The building and use of these hydroelectric generators cause environmental impacts consisting of, but not limited to, disrupting the ecology of rivers involved in the building of said dam, affecting surrounding habitats and ecosystems, and causing soil erosion to the surrounding land. In worst-case scenarios, the dam could break or overflow and cause catastrophic flooding.

This leads us to our last two options: geothermal heating and bioenergy. Geothermal heating involves using the kinetic energy from steam flash formation, similar to wind turbines, to generate electricity. It's often used for heating alone as well, but could be scaled to electricity generation by turning a turbine connected to an axle in a generator. Nevertheless, because of the general scarcity of places in which these special rock formations would be easily accessible, this form of green energy is rarely used.

In the coming years, if investment into this energy source continues, it may be enough to power 10% or more of the world's global energy needs, but for now, it remains a little-used option with very high potential.

This brings us to our last form of green energy. Bioenergy. Bioenergy, as the name suggests, refers to the energy derived from plants and animals, and other forms of organic matter, in other words, biomass. Examples of this include food crop use, crop fuel, wood, and forest waste, as well as the energy harnessed from some unicellular organisms that create this energy through photosynthesis. It is by far the most accessible type of renewable energy, although it cannot be considered a particularly lean one, due to the fact that the burning of these materials often leads to more CO₂ emissions while also destroying ecological land and diminishing biodiversity in favor of mass consumption.

Having looked through all our options, we see now that they each have their pros and cons, which shows us that it would be hard, even impossible, to find a universally fitting source of energy that works in every scenario.

Fortunately for us, there does not need to be.

If we are careful and well-calculated in our use of all renewable energy sources and investment in these resources, we may be able to put the age of fossil fuels behind us in search of a brighter,

cleaner future. Already, we have been seeing countries - China, France, and Brazil are basic examples - in which the use of renewable and green energy far surpasses that of fossil fuels.